Introduction

This literature highlight addresses a central question for our project: **Can optical sensors, specifically Near-Infrared Spectroscopy (NIRS), achieve the phoneme-level temporal resolution and accuracy required for a real-time Silent Speech Interface (SSI)?** To address this, four supporting papers are selected: The first two ([1], [2]) validate the fundamental feasibility of NIRS for laryngeal monitoring, while the latter two ([3], [4]) provide insights into wearable system design and advanced AI processing that can be translated to an NIRS-based platform.

Part I: Validating NIRS for Laryngeal Muscle Sensing

Paper 1: "Development of a Wireless Multichannel Near-Infrared Spectroscopy Sensor System for Monitoring Muscle Activity"

This paper presents a robust, wireless multi-channel NIRS system designed for muscle activity monitoring. The system employs three wavelengths $(750, 805, 850\ nm)$ and two photodiodes per module to calculate relative changes in oxy- and deoxy-haemoglobin $(HbO_2\ and\ Hb)$ concentrations using the Modified Beer-Lambert Law. Key achievements include an excellent signal-to-noise ratio (mean $58\ dB$), low drift (0.12% over 30 minutes), and demonstrated ability to track hemodynamic responses during cuff occlusion and hand motion experiments. The work establishes a high-performance, distributed NIRS sensing paradigm.

Xie et al. provide a foundational hardware blueprint for our project. Their use of multiple wavelengths and source-detector pairs is crucial for obtaining rich, quantifiable physiological data. The system's proven ability to distinguish activation between synergistic muscles supports our idea of decoding complex synergies of laryngeal muscles via NIRS. This paper gives us confidence in the signal quality and hardware architecture needed for our initial goal of simulating and then experimentally validating NIRS sensor placement on the neck.

Paper 2: "AI-boosted and motion-corrected, wireless near-infrared sensing system for continuously monitoring laryngeal muscles"

Liu et al. develop the "LaHMo" platform, a wearable patch that integrates NIRS sensors with an Inertial Measurement Unit (IMU) for continuous laryngeal muscle monitoring. It uses a dual-modality Recurrent Neural Network (RNN), specifically a Gated Recurrent Unit (GRU), to classify physiological events (e.g., swallowing, coughing) by fusing NIRS data with IMU-based motion artifact correction. The system was validated against gold-standard Electromyography (EMG) and through Monte Carlo simulations, confirming its capability to track extrinsic laryngeal muscle activity.

This paper is a direct precedent for applying NIRS to the anterior neck. While LaHMo focuses on classifying gross motor events over seconds, it demonstrates that NIRS signals from the larynx are rich and decodable. The integration of an IMU for motion correction is a design choice we could consider, as neck movements are a major source of noise. Their use of a temporal model (RNN) is also relevant, though for phoneme-level decoding, we will likely require a model architecture with a finer temporal focus. LaHMo bridges the gap between NIRS feasibility and a practical, AI-enhanced wearable system.

Together, [1] and [2] confirm that NIRS can capture robust muscular signals from the larynx. However, the challenge for our project is the slower hemodynamic response of NIRS compared to the millisecond-scale dynamics of phonemes. To overcome this, we hypothesise that a high-density sensor array (inspired by [1]'s multi-channel approach), combined with a predictive, data-driven deep learning model (inspired by [2]'s use of RNNs), can learn the synergistic muscle patterns that precede and constitute speech, effectively mitigating the latency limitation.

Part II: Borrowing from State-of-the-Art SSI System Design

Paper 3: "Machine learning-assisted wearable sensing systems for speech recognition and interaction"

This work presents a Skin-Attached Acoustic Sensor (SAAS) based on a flexible Piezoelectric Micromachined Ultrasonic Transducer (PMUT). The system achieves high-fidelity speech capture in noisy environments (85 dB) and when wearing a mask, with a wide bandwidth (10 Hz–20 kHz) and high sensitivity (-198 dB). Integrated with a convolutional neural network (CNN), it demonstrated >96% accuracy in classifying phonemes and tones, and 99.8% accuracy for full-sentence recognition. The design emphasises wearability, with a flexible, distributed layout and Bluetooth Low Energy (BLE) for wireless operation.

While based on a different sensing modality (mechano-acoustic) and mainly voiced speech, this paper is an exemplary user-centric system design. The seamless wireless integration and demonstration of the device as a versatile controller for HMI (e.g., robot dogs, games) expands our vision beyond a simple speech prosthesis to a general-purpose wearable interface. The use of dimensionality reduction is an essential preprocessing step for simple and interpretable AI/ML models. Furthermore, their use of sliding window and Short-Time Fourier Transforms (STFT) to capture time-frequency features for CNN processing is a powerful preprocessing technique we can learn from for our NIRS data.

Paper 4: "Wearable intelligent throat enables natural speech in stroke patients with dysarthria" (ArXiv preprint)

Tang et al. created an "Intelligent Throat" system using ultrasensitive textile strain sensors to capture laryngeal muscle vibrations and carotid pulse. Its key innovation is a two-stage AI process: a token-level decoding network segments signals into ~100ms tokens for real-time word recognition, and then two Large Language Model (LLM) agents - a Token Synthesis Agent (TSA) and Sentence Expansion Agent (SEA) - synthesise and enrich the output into fluent, emotionally expressive sentences. With the LLM augmentation, the system achieved a WER of 4.2% and increased user satisfaction by 55%.

This paper provides a compelling roadmap for our AI pipeline. The token-level approach is crucial for real-time, continuous operation. The use of Knowledge Distillation to reduce their model's computational load by 75.6% is a pivotal strategy we will consider for edge deployment. Most importantly, the post-decoding use of an LLM agent is a powerful idea we can adopt. By focusing our core model on the **bottleneck task of accurate phoneme-to-word decoding from NIRS signals**, we can offload the tasks of grammatical correction, contextual enrichment, and even multi-language translation to a pre-existing LLM, enhancing the naturalness and utility of the final output.

Papers [3] and [4] demonstrate that with high-quality signal acquisition and a sophisticated AI pipeline, highly accurate and natural SSI is achievable. From [3], we learn the importance of wearability, wireless connectivity, and versatile HMI. From [4], we learn the power of a segmented AI approach, combining a specialised decoder with a general-purpose LLM for high-quality output, and the model optimisation techniques like Knowledge Distillation for practical deployment.

Conclusion

Papers [1] and [2] provide the scientific and technical foundation for using NIRS on the larynx, while indicating the challenge of achieving phoneme-level speed. Papers [3] and [4] provide the engineering and architectural reference to overcome this challenge: high-density, comfortable hardware, real-time tokenisation, Knowledge Distillation, and LLM post-processing. By integrating the NIRS with AI and user-centred design, our project is positioned to develop a discreet, neck-worn device that can seamlessly translate laryngeal movements into natural speech.

Citations

[1] F. Xie, S. Huang, T. Miao, S. He, Z. Lin, and L. Xie, “Development of a Wireless Multichannel Near-Infrared Spectroscopy Sensor System for Monitoring Muscle Activity,” *IEEE Sensors Journal*, vol. 22, no. 23, pp. 22714–22724, Nov. 2022, doi: https://doi.org/10.1109/jsen.2022.3216351.

[2] Y. Liu *et al.*, “AI-boosted and motion-corrected, wireless near-infrared sensing system for continuously monitoring laryngeal muscles,” *Proceedings of the National Academy of Sciences*, vol. 121, no. 51, Dec. 2024, doi: https://doi.org/10.1073/pnas.2410750121.

[3] T. Liu *et al.*, “Machine learning-assisted wearable sensing systems for speech recognition and interaction,” *Nature Communications*, vol. 16, no. 1, Mar. 2025, doi: https://doi.org/10.1038/s41467-025-57629-5.

[4] C. Tang *et al.*, “Wearable intelligent throat enables natural speech in stroke patients with dysarthria,” *arXiv (Cornell University)*, Nov. 2024, doi: https://doi.org/10.48550/arxiv.2411.18266.